

Composition Pair B1.14 + A1.07 — Directed Selection and Path Retraceability: The Reinforcing Pair Where Governance Acts Require Accountability Records and Accountability Records Require Governed Acts

Author: Wenxin Li (Independent Researcher)

ORCID: 0009-0004-8065-3235

Date: May 12, 2026

License: Creative Commons Attribution 4.0 International (CC BY 4.0)

Attribution

This work derives from and formalizes the instinct/reasoning separation pattern introduced in *The Instinct/Reasoning Separation Outside the Model* (Li, April 2026), the second paper in the CKS theory series following *Coordination Outside the Model: A Human-Governed Substrate Pattern for AI Systems* (Li, April 2026). It does not introduce new axioms. Its sole contribution is to articulate, in operational form, the reinforcing composition relationship between directed selection (B1.14) and path retraceability (A1.07) — a pair whose mutual dependency constitutes the deployment's governance history and makes that history auditable.

Abstract

Among the composition pairs that describe how the architectural commitments of CKS Paper 2 jointly operate, the B1.14 + A1.07 pair is structurally distinctive: the relationship is reinforcing rather than merely enabling. Directed selection (B1.14) produces the primary governed events that constitute a deployment's governance history — each human decision to modify entity DNA is a governance act authorized under Paper 1's standard authority architecture. Path retraceability (A1.07) provides the accountability architecture through which those governance acts are recorded, verified, and made demonstrable. The reinforcement runs in both directions: B1.14 requires A1.07 because an unrecorded governance act is a governance act without accountability; A1.07 requires B1.14 because a retraceable substrate with no governed events in its history records what the deployment did operationally but not what governance decided about how it should behave. The DNA version chain (B2.69) is the operational embodiment of the pair — each DNA version is a directed selection event carrying A2.40 provenance, and the chain of versions constitutes the deployment's retraceable governance decision history. A6.02's retroactivity commitment becomes verifiable through this pair because modification timestamps and operation timestamps together determine which DNA version governed which operation. The pair's failure modes are Ungoverned DNA Modification (B3.15) when B1.14's recording requirement is violated, and Provenance Void (B3.23) when A1.07 infrastructure is systematically absent.

1. Pair identification and the nature of the reinforcing relationship

Pair: B1.14 (directed selection) + A1.07 (path retraceability).

Relationship type: Reinforcing pair — each commitment requires the other for its constitutive purpose to be satisfied.

A reinforcing pair differs from an enabling pair. In an enabling pair, one commitment provides capabilities that another needs in order to function. In a reinforcing pair, neither commitment's constitutive purpose is satisfiable without the other: B1.14's constitutive purpose is governed DNA modification — not merely modification, but modification under an authority architecture that makes governance acts verifiable; A1.07's constitutive purpose is retraceable governance history — not merely a record of events, but a record through which human governance decisions can be reconstructed and audited. Neither purpose is satisfiable without the other. An unrecorded governance act satisfies modification but not governed modification. A retraceable record of operational events satisfies traceability but not governance history.

The reinforcing relationship operates at the level of what each commitment is *for*, not merely at the level of what each commitment technically requires.

2. B1.14 directed selection — the governance act that constitutes the deployment's governance history

Directed selection in CKS Paper 2 names the mechanism by which a deployment's DNA evolves under human governance. A directed selection event is a human decision to modify entity DNA — specifically, to modify the orchestration substrates, behavior substrates, schemas, and rules that constitute the DNA layer of a cell, aspect, or Self. The selection is directed because the criteria for which DNA changes are accepted, retained, or reverted are themselves defined by human governance: not goal-defined-by-internal-system-state, not goal-defined-by-fitness-function, not goal-defined-by-emergent-bias (§7.2), but goal-defined-by-human-governance, with selection criteria as substrate content.

What makes a directed selection event a governance act rather than merely an edit is Paper 1's standard authority architecture: authorization per A2.47 (the human exercising a modification right conferred by the authority structure), authorship per A2.04 (the modification is attributed to an identified human acting under identified authority), and recording per A2.40 (the modification is written into the substrate with the six provenance metadata fields). These three requirements together are what make the selection directed in the governance sense: not merely selected toward a goal, but selected under an authority architecture that makes the selection verifiable, attributable, and accountable.

Directed selection events are the primary *governed events* in the deployment's governance history. Other substrate events — cell executions, LLM outputs, task instances recorded in the action layer — are also recorded in the substrate, but they are operational events rather than governance

decisions. A directed selection event records a human governance decision: a decision about how the entity *should behave*, implemented as a change to the DNA that governs its behavior. The deployment's governance history is substantially constituted by its directed selection history.

This point has a structural consequence: A1.07 path retraceability is architecturally valuable in proportion to the density of governed events in the retraceable record. A deployment that practices directed selection has a governance history that retraceability makes auditable. A deployment that modifies DNA without directed selection governance has a record that retraceability makes legible but not auditable in the governance sense — because what is recorded are edits without governance acts.

3. A1.07 path retraceability — the accountability architecture that makes governed acts verifiable

Path retraceability is the property that any decision, output, or piece of substrate content can be traced back through a path of antecedent substrate content to the inputs and orchestration rules that produced it (A1.07; Rajabi and Kafaie, 2022). For the tracing to be possible from substrate content alone — without consulting external logs, agent memory, or human recollection — each piece of substrate content must carry sufficient provenance. The six provenance metadata fields A2.40 specifies accomplish this: writer attribution, timestamp, antecedent reference, rule reference, rationale where required, and contradiction relationship where applicable.

In the context of directed selection events, A1.07 retraceability has a specific and critical function: it is the accountability architecture through which governance acts can be verified. A governance act that is authorized, authored, and recorded with complete A2.40 provenance is a governance act that can be verified — the verifier can answer who authorized it, who authored it, when it was made, under what orchestration rule, drawing on what antecedent substrate content, for what stated rationale. Without A1.07's retraceability infrastructure, governance acts are performed but not verifiable: the modification was made, but the governance chain cannot be reconstructed from substrate content.

This is the precise sense in which A1.07 is required by B1.14: a directed selection event that is not retraceable is a governance act without accountability. The authority-not-labor architecture (A1.01) requires that governance acts be verifiable, not merely that they be performed. Verifiability requires retraceability. Retraceability requires A2.40 provenance. A2.40 provenance requires A1.07 infrastructure.

Conversely, A1.07 retraceability has *content* in a governance history only to the extent that directed selection is practiced as a governance discipline. Path retraceability makes auditable whatever governed events the substrate records. If the substrate records governed DNA modification events, the retraceable history includes governance decisions. If the substrate records only operational events, the retraceable history shows what the deployment did but not what governance decided.

The governance history dimension of retraceability depends on B1.14 being practiced.

4. The reinforcing relationship: how each direction of dependency operates

Direction 1 — B1.14 requires A1.07 for governance legitimacy.

A directed selection event derives its legitimacy as a governance act from the three requirements named in §2: authorization, authorship, and recording. Recording is the operative term here — the modification must be written into the substrate with A2.40 provenance. Recording with A2.40 provenance is the instantiation of A1.07 retraceability at the moment of the governance act. If A1.07 infrastructure is absent — if the substrate schema does not require provenance fields, if orchestration rules do not enforce their population — then directed selection events may be authorized and authored but are not recorded with sufficient provenance for retraceability. The governance act is performed but not retraceable. An untraceable governance act is a governance act that cannot be verified. A governance act that cannot be verified is governance in name only: the human made a decision, but no one can demonstrate this from the substrate.

The A5.09 four accountability questions — who decided, what was decided, when it was decided, why it was decided — are answerable from substrate content only if A1.07's provenance fields are present at the moment of each directed selection event. B1.14's governance requirement is satisfied only when A1.07 is co-present.

Direction 2 — A1.07 requires B1.14 for meaningful governance history content.

Path retraceability is an architectural property of the substrate. The property can be satisfied even if the deployment has no governance practice at all, provided the substrate schema and orchestration rules enforce A2.40 provenance on whatever events are recorded.

But retraceability over a substrate whose events are all operational — task executions, LLM outputs, action-layer records — is retraceability over an execution history, not a governance history. The retraceable record answers 'what did the deployment do' but not 'what did governance decide about how it should behave.' A deployment auditor examining such a substrate can trace every operational action but cannot verify that the DNA content governing those actions arrived through human governance decisions, because no directed selection events appear in the record. DNA modifications may have been made informally, autonomously, or without authorization — and the retraceable record cannot distinguish these cases.

B1.14 directed selection is what puts governance decisions into the retraceable record. When directed selection is practiced as the governance discipline for DNA modification, the governance history dimension of retraceability is populated. The auditor can trace not only what the deployment did but what governance decided about how it should behave, and when those decisions were made.

5. The DNA version chain as operational embodiment of the pair

The DNA version chain (B2.69) is the artifact in which B1.14 + A1.07 compose operationally. Each DNA version in the chain is the result of a directed selection event (B1.14) — a human governance decision to modify entity DNA, made under the authority architecture. Each DNA version record carries A2.40 provenance (A1.07) — writer attribution, timestamp, antecedent reference, rule reference, rationale, and contradiction relationship as applicable. The chain of versions, ordered by modification timestamp, constitutes the complete retraceable governance decision history of the entity's DNA evolution.

The version chain is not merely a version history in the technical sense. Version history records what each successive state was; the DNA version chain under B1.14 + A1.07 records *why and under what authority* each version exists. The distinction between version history and version chain with governance provenance is the distinction that makes compliance audit possible: the auditor does not merely see that the DNA changed, but can reconstruct the complete governance chain — every authorization, every authorship attribution, every rationale — from substrate content alone.

The A6.02 retroactivity connection.

A6.02 commits that new DNA versions govern behavior from the modification timestamp forward, not retroactively. This commitment makes the version chain's timestamps structurally significant: each version has a validity window defined by its own modification timestamp and the timestamp of the succeeding version. For any operation recorded in the substrate with an operation timestamp, the DNA version that governed that operation is determinable by comparing the operation timestamp against the version chain's modification timestamps.

This determinability is what makes A6.02 compliance *verifiable*. Without modification timestamps — without A1.07's timestamp field as part of A2.40 provenance — the question 'which DNA version governed this operation' cannot be answered from substrate content alone. With modification timestamps present in the version chain, A6.02 compliance is auditable: the verifier reads the version chain, identifies the version whose validity window contains the operation timestamp, and that version is the DNA that governed the operation. The pair together — B1.14 producing the version chain, A1.07 providing the timestamps within it — enables A6.02 compliance verification.

6. Co-presence requirement and failure modes

B1.14 without A1.07 (directed selection without retraceability infrastructure).

DNA modifications are made under governance authorization — humans authorize and author the changes — but the substrate schema and orchestration rules do not enforce A2.40 provenance. The resulting modifications are unauthenticated in the substrate: the DNA may have changed, but the modification record does not carry sufficient provenance for retraceability. The governance history cannot be reconstructed from substrate content.

The operational failure is compliance demonstration: when an auditor asks whether current DNA content arrived through authorized directed selection, the substrate cannot answer from its own content. The governance acts were performed but are unverifiable. B3.15 (Ungoverned DNA Modification) names the anti-pattern in its recording-absent form: DNA modification without retraceable provenance, regardless of whether the modification was authorized in a non-substrate process.

A1.07 without B1.14 (retraceability without directed selection governance practice).

The substrate faithfully records events with A2.40 provenance — writer attribution, timestamps, antecedent references, rule references all present on every piece of substrate content. The substrate is retraceable. But no directed selection governance practice exists: DNA modifications are made outside the governed directed selection framework — informally, autonomously, or through channels that do not require the authority architecture. The retraceable record contains operational events but no governance decisions.

The operational failure is governance history: the retraceable substrate cannot support a compliance claim that DNA content was arrived at through human governance decisions. The auditor can trace every operational action but not a single governance act over DNA content. This is a governance-history void even on a perfectly retraceable substrate. B3.23 (Provenance Void) names the anti-pattern when A1.07 infrastructure is systematically absent; the symmetric failure here — retraceability present but no governed events to trace — is a complementary failure mode the pair's framing makes distinguishable.

Full co-presence (B1.14 + A1.07 together).

Every directed selection event creates a retraceable governance record: the modification is authorized per A2.47, authored per A2.04, and recorded per A2.40 with full provenance. The DNA version chain per B2.69 grows with each governance decision, and the version chain is the retraceable governance history. Compliance audit traces the version chain from any current DNA content back through its complete directed selection history to its origin. A6.02 retroactivity is verifiable through the modification timestamps. The A5.09 four accountability questions are answerable for every governance act from substrate content alone.

7. Detection and operational integration

Detection. Three checks jointly verify pair completeness.

B2.72 directed selection verification: Are all DNA modifications authorized and recorded? A verification pass over the DNA version chain checks that every version has complete A2.40 provenance and that the provenance attributes the modification to a human acting under A2.47 authorization. A version without full provenance fails the directed selection verification and signals a B3.15 or B3.23 condition.

A5.08 provenance-completeness test: Are all modification records complete? The provenance-completeness test checks that every DNA modification record in the substrate carries all six A2.40 fields — not merely that a record exists, but that the record carries enough provenance for retraceability. An incomplete record satisfies recording but not retraceability.

B2.69 DNA version chain completeness: Does the version chain provide a complete retraceable governance history? The chain is complete if and only if (a) every modification to the DNA is represented as a version in the chain, (b) every version carries A2.40 provenance, and (c) the chain is unbroken from the entity's origin to its current state. A broken chain — a version without a retraceable antecedent — signals that some modification was made outside the governed directed selection framework.

Pair completeness: Can any current DNA content be traced through authorized directed selection events to its origin? This is the composite test: it is satisfied only when all three sub-checks pass for all versions in the chain.

Operational integration.

The governance cycle under the fully co-present pair operates as follows. A human makes a governance decision to modify entity DNA per B1.14. The modification is authorized per A2.47 and attributed to the authorizing human. The human or an LLM operating under human direction drafts the modification per A2.04. The draft is reviewed and accepted under whatever governance shape Paper 2 §8 specifies for the mechanism. On acceptance, a new DNA version is written per B2.69 with a modification record carrying all six A2.40 fields. The new version's timestamp is the moment from which it governs behavior per A6.02. Subsequent operations that encounter the entity read the current DNA version, and the operation timestamp together with the version chain determines unambiguously which version was operative per the A6.02 retroactivity rule. Compliance audit uses A1.07 retraceability to walk the version chain, verifying governance acts for each version and answering the A5.09 accountability questions from substrate content alone.

8. Conclusion

The B1.14 + A1.07 reinforcing pair is the governance-history backbone of the deployment. Directed selection puts human governance decisions into the substrate as versioned DNA modifications; path retraceability provides the provenance infrastructure that makes those decisions verifiable, attributable, and auditable. Neither commitment satisfies its constitutive purpose without the other: governance acts without retraceability are unaccountable; retraceability without governance acts traces operations but not decisions. The DNA version chain (B2.69) is where the pair composes into an artifact — each version a governance decision, each record a provenance trace, the chain as a whole the deployment's auditable governance history. A6.02 retroactivity is the downstream property the pair makes verifiable. B3.15 and B3.23 are the failure modes that activate when either commitment is absent.

Source papers

Li, W. (2026). *Coordination Outside the Model: A Human-Governed Substrate Pattern for AI Systems*. April 2026. ORCID: 0009-0004-8065-3235.

Li, W. (2026). *The Instinct/Reasoning Separation Outside the Model: Extending the Coordination Knowledge Substrate Pattern to AI Selves under Human Governance*. April 2026. ORCID: 0009-0004-8065-3235.

How to cite this note

Li, W. (2026). *Composition Pair B1.14 + A1.07 — Directed Selection and Path Retracement: The Reinforcing Pair Where Governance Acts Require Accountability Records and Accountability Records Require Governed Acts*. May 12, 2026. ORCID: 0009-0004-8065-3235.